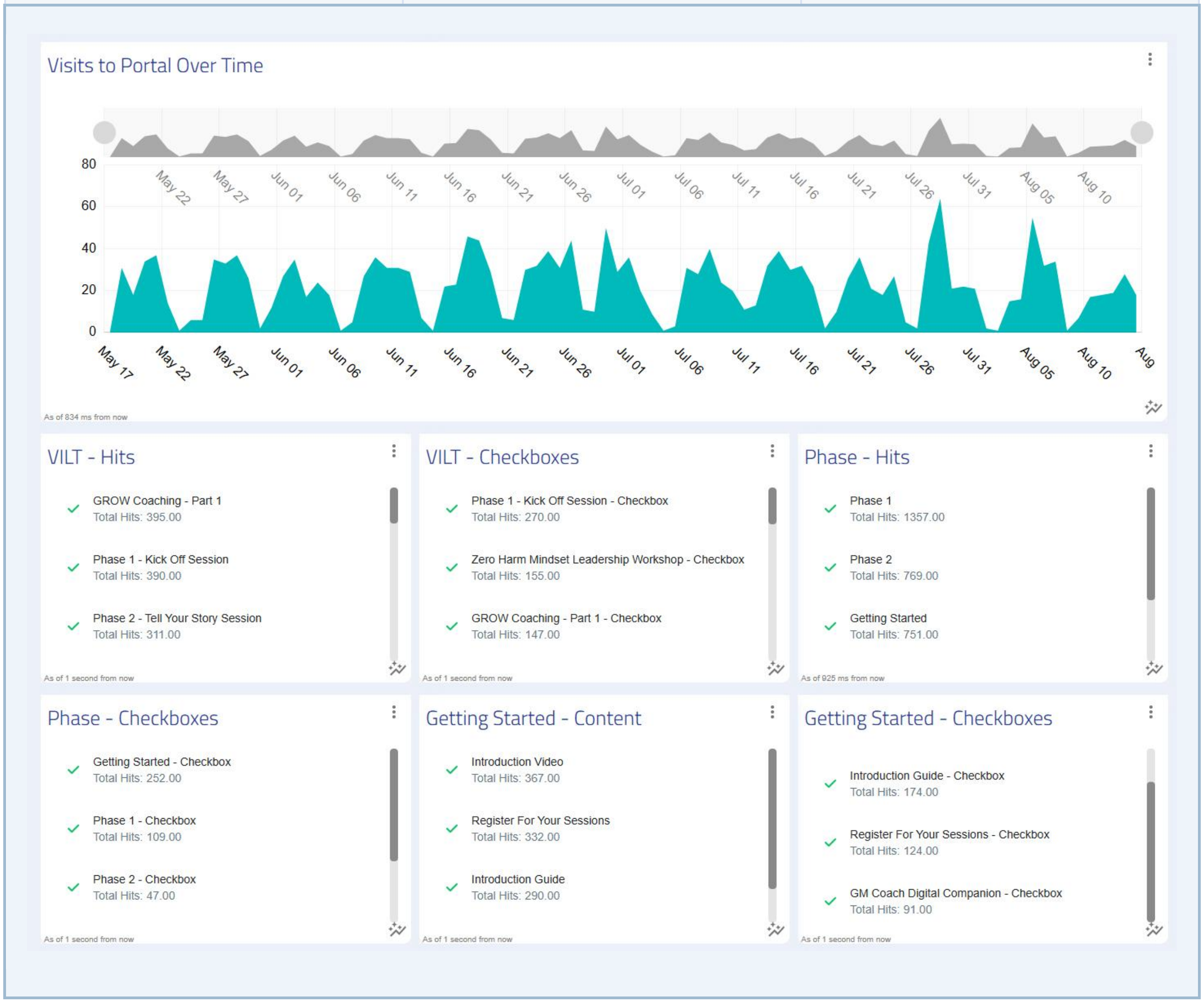
CASE STUDY

xAPI / LRS Implementation

Creating a learning data foundation for portal tracking and reporting

ROLE	PRIMARY TOOLS	FOCUS
Learning Technologist / Learning Experience Developer / eLearning Developer / Web Designer	**xAPI • LRS (Veracity) • Storyline • JavaScript**	**Tracking • Analytics • Portal Integration**



The Challenge

The learning portals created a need for better visibility into learner activity and progress. I wanted a way to capture meaningful learner interactions, persist completion data, and turn that activity into useful reporting. This led me to learn about xAPI and Learning Record Stores (LRSs) and explore how they could extend what the learning portals could do.

Learning About xAPI and the LRS

I learned how xAPI could capture learner activity outside the traditional LMS and how an LRS could store that activity for later analysis. I applied that knowledge to the learning portals, designing an approach in which portal activity could be recorded, stored, and retrieved as needed.

I then implemented xAPI and an LRS from Veracity as part of the portal architecture. Storyline provided the learner-facing experience, JavaScript extended the functionality, xAPI captured activity, and the LRS provided a central location for storing and reporting on learner data.

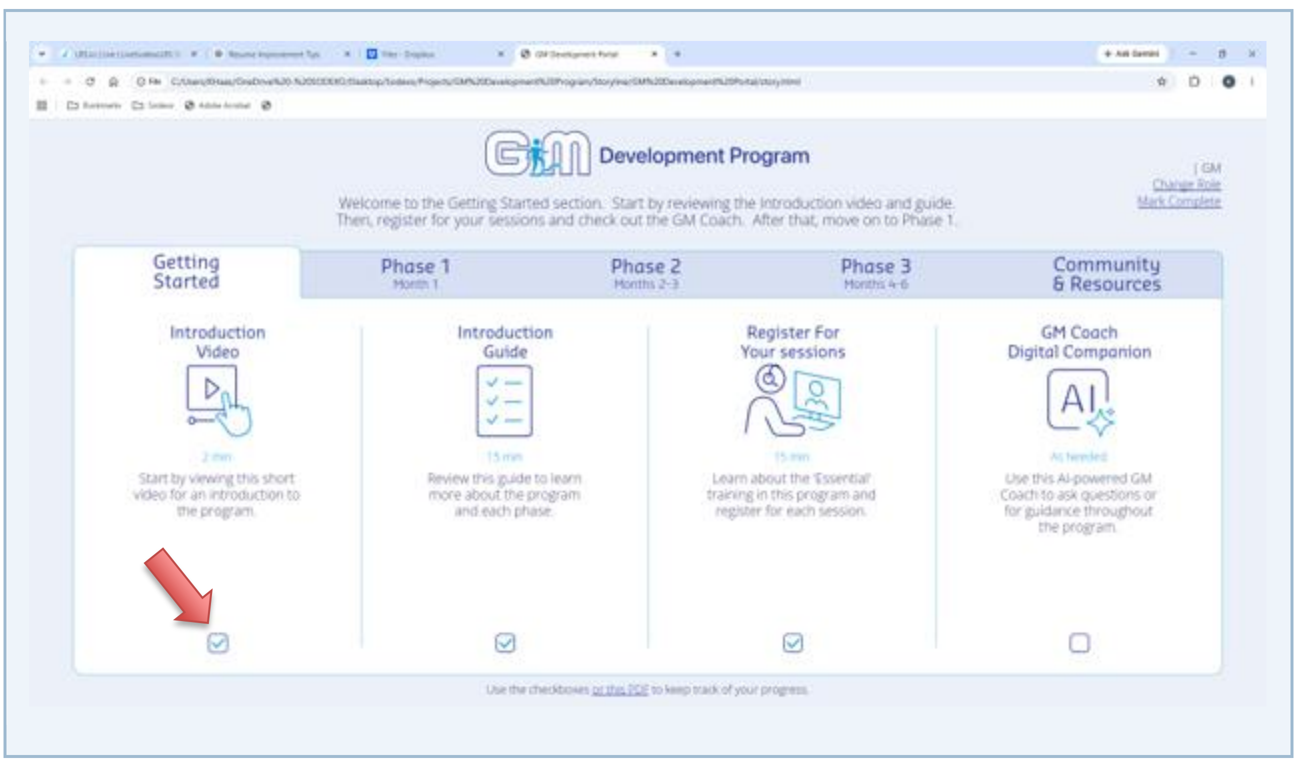
Implementation & Integration

xAPI Used to capture learner activity from the learning portals and communicate that activity to the LRS.	**LEARNING RECORD STORE (LRS)** Provided the central data store for learner activity and supported reporting and dashboard development.
ARTICULATE STORYLINE Provided the learner-facing portal experience and worked with xAPI, the LRS, and JavaScript to support persistent progress.	**JAVASCRIPT** Extended Storyline functionality, retrieved learner completion data from the LRS to maintain progress between browser sessions.

Portal Progress Tracking

One of the most visible applications was persistent learner progress. Learners could visually identify which training activities they had completed using interactive checkboxes within the portal. Completion information was stored in the LRS and retrieved through JavaScript when learners returned to the portal, allowing their progress to persist across browser sessions.

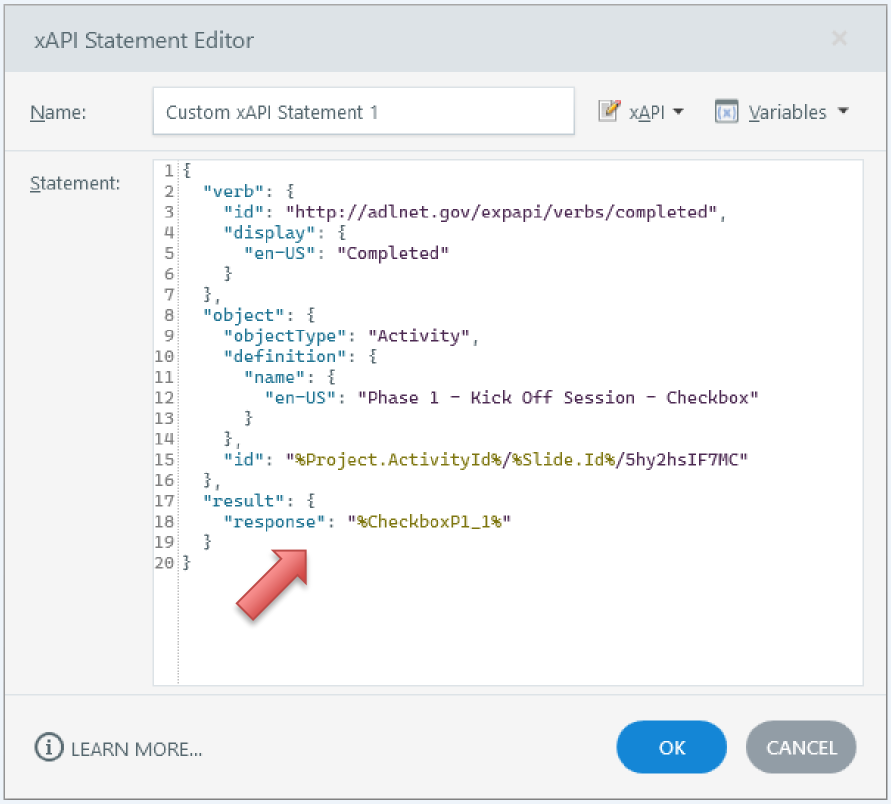
This allowed the portals to function as more than collections of links. They became interactive learning experiences with persistent progress tracking and a data layer that could be used for analysis and reporting.

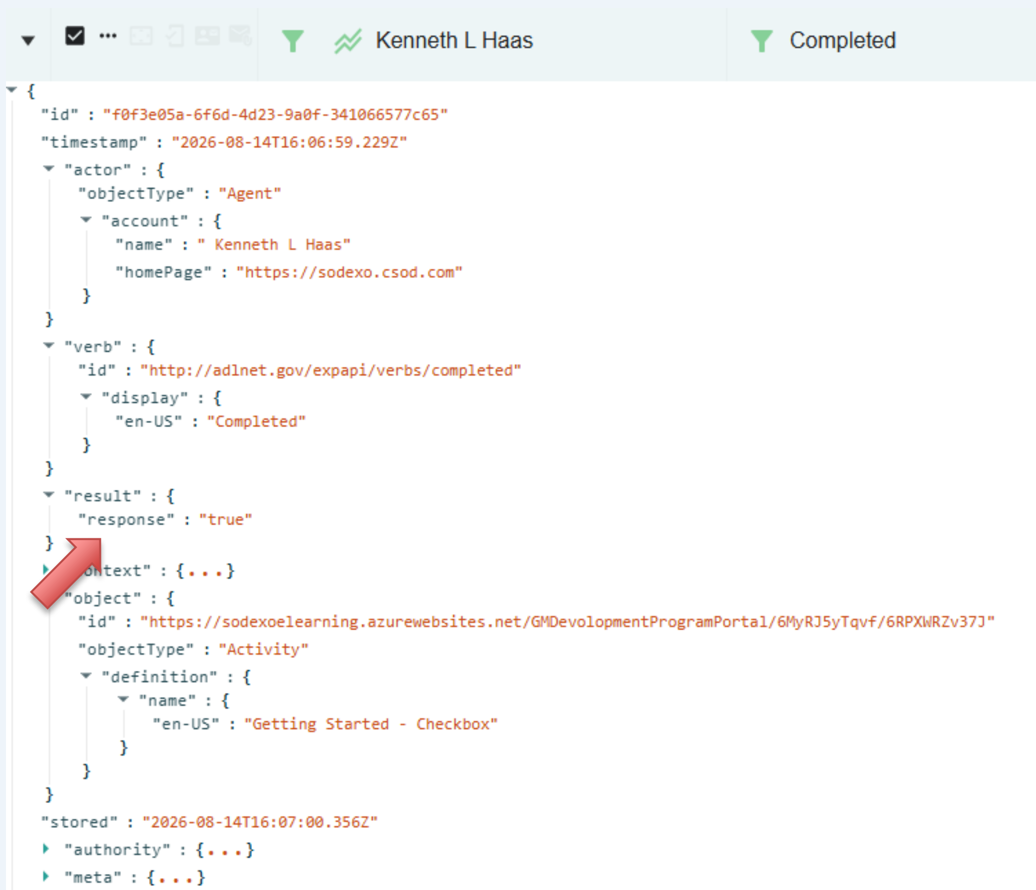


Behind the Experience

LEARNER	STORYLINE PORTAL	xAPI	LRS (Veracity)	JAVASCRIPT → STORYLINE

Conceptual flow: learner activity is captured through the portal, recorded through xAPI in the LRS, and retrieved by JavaScript so the Storyline experience can reflect persistent progress.




Kenneth L Haas
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}
}
}
"stored" : "2026-08-14T16:07:00.356Z"
"authority" : {...}
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JavaScript Editor
Object reference
But_P1B1
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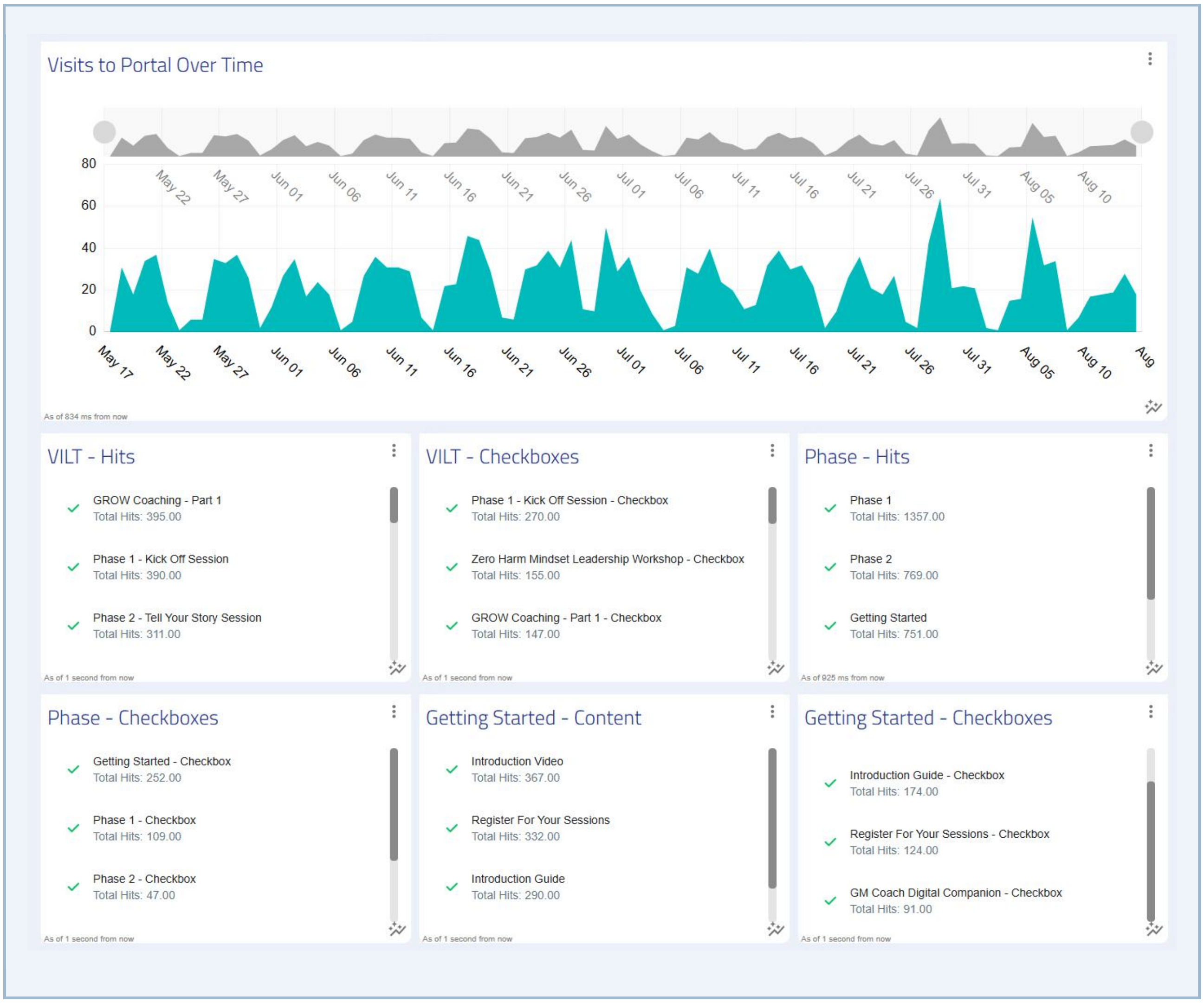
{ objID: "6MyRJ5yTqvf/5lWnPCsfuwQ", varName: "CheckboxP1_2" },
{ objID: "6MyRJ5yTqvf/6amPca9MjI3", varName: "CheckboxP1_3" },
{ objID: "6MyRJ5yTqvf/6VXtFXG9OMO", varName: "CheckboxP1_4" }
];
for (const item of checkboxMap) {
try {
await getCheckbox(item.objID, item.varName);
} catch (xhr) {
console.error("Error fetching checkbox for", item.varName, xhr);
}
}
})();
LEARN MORE...
OK
CANCEL

Reporting & Analytics

I created custom reporting dashboards in the LRS to make learner activity more visible and useful. The dashboards provided a way to review activity captured from the portals and analyze learner engagement and progress across the learning programs.



Business Impact

- Established a repeatable xAPI and LRS approach for capturing and storing learner activity from custom learning portals.
- Created custom LRS reporting dashboards to analyze learner activity and program engagement.
- Implemented persistent learner progress tracking so completion status could be retained across browser sessions.
- Extended the capabilities of Articulate Storyline by combining JavaScript, xAPI, and LRS technology.

What I Learned

Learning xAPI and LRS technology expanded my capabilities beyond traditional eLearning development and helped me think about learning experiences as connected systems rather than standalone courses. The work gave me hands-on experience implementing learning data capture, persistence, and reporting within a custom digital learning environment.